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Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Time: 03 Hours

Code: 103404 ***Subject: Signals and Systems*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven questions only): **[14]**

(a) If $X(f)$ is the Fourier transform of $x(t)$, then what is the Fourier transform of $x(2t)$?
(i) $X(f/2)$
(ii) $0.5X(f/2)$
(iii) $2X(2f)$
(iv) $X(f)$

(b) Which of the following statements is true?
(i) Discrete-time signals are always periodic
(ii) Continuous-time signals cannot be periodic
(iii) Discrete-time signals can be periodic or aperiodic
(iv) Continuous-time signals are always aperiodic

(c) Find the fundamental period of $x(t) = \sin(5\pi t) + \cos(7\pi t)$.
(i) 0.2 sec
(ii) 0.4 sec
(iii) 0.8 sec
(iv) 1.0 sec

(d) If a system's impulse response is $h(t) = \delta(t) + \delta(t-1)$, the system is:
(i) Causal
(ii) Anti-causal
(iii) Non-causal
(iv) Time-invariant

(e) Which system property ensures that the output depends only on past and present inputs?
(i) Linearity
(ii) Causality
(iii) Stability
(iv) Time-invariance

(f) Which of the following is an example of a deterministic signal?
(i) White noise
(ii) Sinusoidal signal
(iii) Random signal
(iv) Gaussian noise

(g) The impulse response of an LTI system is given by $h(t) = e^{-2t}u(t)$. Find the system response to an input $x(t) = e^{-t}u(t)$.
(i) $[e^{-t}-e^{-2t}] / [3]$
(ii) $[e^{-t}-e^{-2t}] / [2]$
(iii) $e^{-t}- e^{-2t}$
(iv) $[e^{-t}-e^{-2t}] / [e^{-3t}]$

(h) The step response of an LTI system is obtained by:
(i) Taking the derivative of the impulse response
(ii) Integrating the impulse response
(iii) Taking the Fourier transform of the impulse response
(iv) Adding two impulse responses

(i) The Laplace transform of $e^{-at}u(t)$ is:
(i) $[1] / [s+a]$
(ii) $[1] / [s-a]$
(iii) $[s] / [s+a]$
(iv) $[s] / [s-a]$

(j) If a signal contains frequency components up to 5 kHz, what should be the minimum sampling rate?
(i) 2.5 kHz
(ii) 5 kHz
(iii) 10 kHz
(iv) 20 kHz

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Q.2 Solve both questions : **[14]**

(a) Find the inverse Laplace transform of: F(s) =[s(s+2)] / [(s+1)(s+3)(s+5)].

(b) Given a system with a difference equation $y[n] - 0.5y[n-1] = x[n]$, determine its stability.

Q.3 Solve both questions : **[14]**

(a) Find the state space model of the system represented by the differential equation: $d^3y/dt^3 + 8d^2y/dt^2 + 9dy/dt = 3x(t)$.

(b) A system is described by the equation: $y(t) = 3x(t) + 2$. Check whether the system is linear or not.

Q.4 Solve both questions : **[14]**

(a) Compute the poles and zeros of F(s) =[s+1] / [$s^3 + 7s^2 + 10s + 18$].

(b) Consider a system defined by $y(t) = tx(t)$. Determine whether it is linear and shift-invariant.

Q.5 Solve both questions : **[14]**

(a) Compute the output $y[n]$ of an LTI system with impulse response h[n] ={1, 2, 1} and input $x[n]$ ={2, 1, 3} using convolution.

(b) Compute the energy and power of the signal $x(t) = e^{-t}u(t)$.

Q.6 Solve both questions : **[14]**

(a) Find the Fourier transform of $x(t) = e^{-2|t|}$.

(b) Find the Laplace Transform of $x(t) = e^{-t}u(t)$ and determine its region of convergence.

Q.7 Solve both questions : **[14]**

(a) Given $x(t)$ =sin(10t) +cos(15t), determine whether it is periodic and find the fundamental period.

(b) Compute the z-transform of $x[n] = -b^n u[n-1]$.

Q.8 Solve both questions : **[14]**

(a) Find the step response of a system with impulse response h(t) = $e^{-3t}u(t)$.

(b) Determine the even and odd components of the signal $x(t) = t^2 + 3t + 2$.

Q.9 Write short notes on any two of the following: **[14]**

(a) Discuss the properties of a system with respect to linearity, causality, and stability.

(b) Define Fourier series and explain its importance in signal representation.

(c) State and explain the Sampling Theorem and its significance.

(d) State and explain Parseval's Theorem with its application.

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